LAB 6

TASK 1

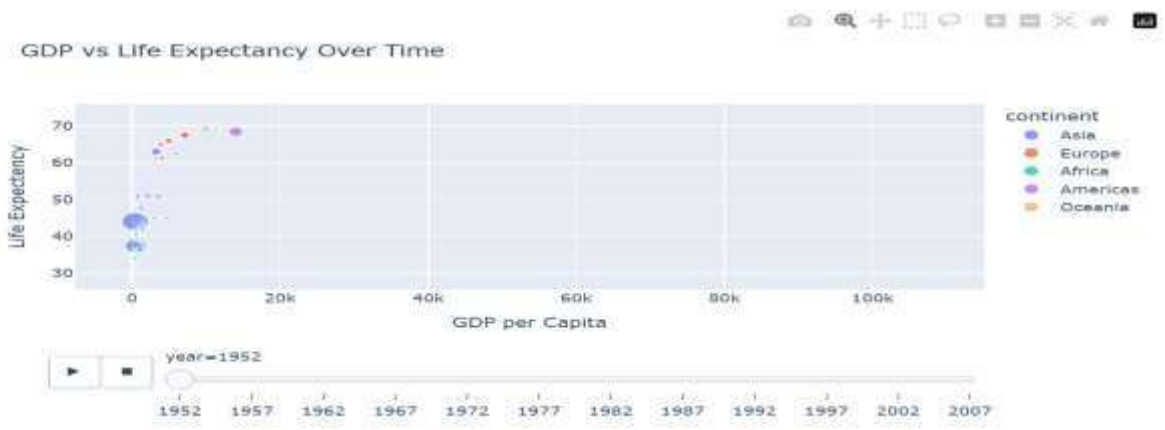
Scatter Plot with Slider (Animation)

- Plot GDP per capita (gdpPercap) vs life expectancy (lifeExp) for all countries.
- Use population (pop) to size the points.
- Color points by continent.
- Add a slider to move through different years.
- Title: "GDP vs Life Expectancy Over Time".
- Label the x-axis as "GDP per Capita (USD)" and the y-axis as "Life Expectancy (Years)".
- Ensure hover tooltips display country names.

INPUT

```
import plotly.express as px data

= px.data.gapminder()

data.head()


fig = px.scatter(data, x="gdpPercap", y="lifeExp", title='GDP vs Life Expectancy Over Time',size="pop", labels={'gdpPercap': "GDP per Capita", 'lifeExp': "Life Expectency"}, animation_frame='year', hover_name='country', color='continent')

fig.show()
```

OUTPUT



TASK 2

Line Plot with Dropdown Selector

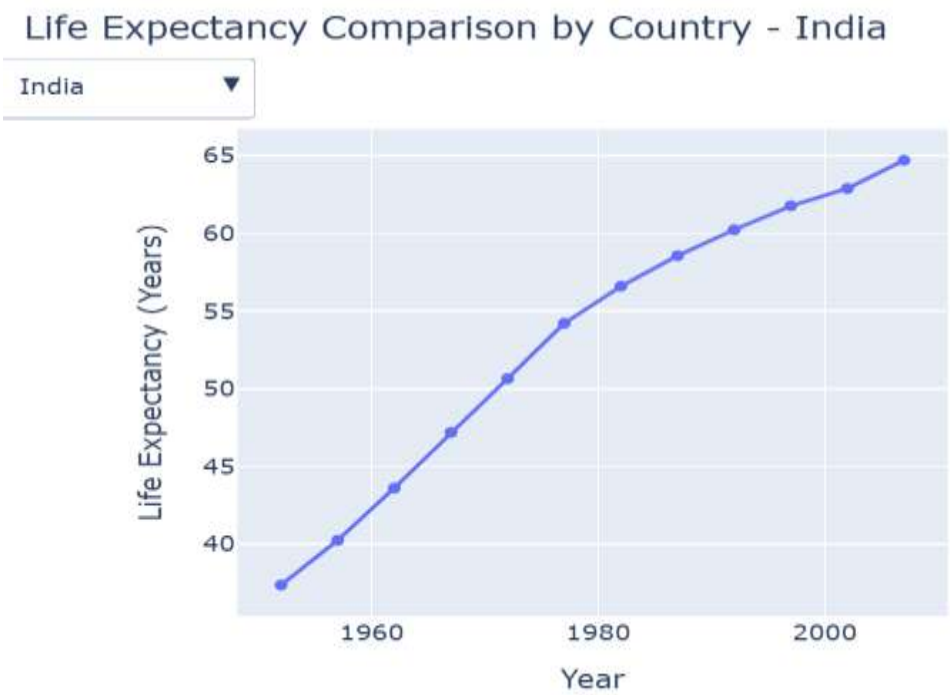
- Plot life expectancy (lifeExp) vs year (year) for multiple countries (e.g., India, China, USA, Brazil).
- Add a dropdown menu to select which country's line is visible.
- Use distinct line colors for each country.
- Title: "Life Expectancy Comparison by Country".
- Label the x-axis as "Year" and the y-axis as "Life Expectancy (Years)".
- Place the legend at the bottom.

INPUT

```
 df_countries = data[data.country.isin(["India", "China", "United States",

"Brazil"])]


fig = go.Figure()


colors = ['red', 'blue', 'green', 'orange']


countries = ["India", "China", "United States", "Brazil"]


for i, country in enumerate(countries):

    df_c = df_countries[df_countries.country == country]

    fig.add_trace(go.Scatter(x=df_c.year, y=df_c.lifeExp, mode='lines',

name=country, line=dict(color=colors[i]), visible=(i==0)))


buttons = []


for i, country in enumerate(countries):
```

```
    visible = [j==i for j in range(4)]
    buttons.append(dict(label=country, method="update", args=[{"visible":
visible}, {"title": f"Life Expectancy for {country}"}]))


fig.update_layout(updatemenus=[dict(type="dropdown", active=0,
buttons=buttons, x=0.1, y=1.1, xanchor="left", yanchor="top")], title="Life
Expectancy Comparison by Country", xaxis_title="Year", yaxis_title="Life
Expectancy (Years)")
```

OUTPUT



TASK 3

Bar Plot with Buttons (Switch Between Variables)

- Plot population (pop) by continent for 2007.
- Add buttons to toggle between population and GDP per capita (gdpPercap).
- Use grouped bars for clarity.
- Title: "Continent Comparison (2007)".
- Label the x-axis as "Continent" and the y-axis as "Value".

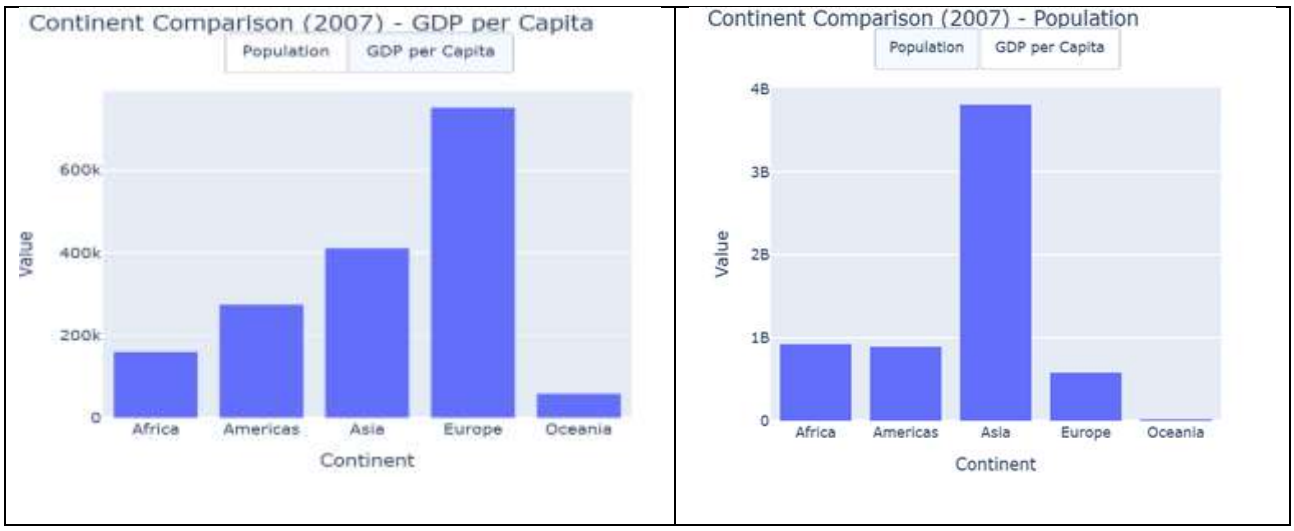
- Display the legend at the top-right corner.

INPUT

```
import plotly.graph_objects as go


df_2007 = data[data.year == 2007]

df_2007_continent_pop =

df_2007.groupby("continent")["pop"].sum().reset_index()


df_2007_continent_gdp =

df_2007.groupby("continent")["gdpPercap"].mean().reset_index()


fig = go.Figure()


fig.add_trace(go.Bar(x=df_2007_continent_pop.continent,

y=df_2007_continent_pop["pop"], name="Population"))

fig.add_trace(go.Bar(x=df_2007_continent_gdp.continent,

y=df_2007_continent_gdp["gdpPercap"], name="GDP per Capita", visible=False))


buttons = [dict(label="Population", method="update", args=[{"visible": [True,

False]}, {"title": "Continent Comparison (2007)", "yaxis": {"title":

"Value"}}]),

        dict(label="GDP per Capita", method="update", args=[{"visible":

[False, True]}, {"title": "Continent Comparison (2007)", "yaxis": {"title":

"Value"}}])]
```

```
fig.update_layout(updatemenus=[dict(type="buttons", buttons=buttons)],
title="Continent Comparison (2007)", xaxis_title="Continent",
yaxis_title="Value", legend=dict(x=0.8, y=1))


fig.show()
```

OUTPUT



TASK 4

Map with Projection Selector

- Plot life expectancy (lifeExp) by country on a world map for 2007.
- Use population (pop) to size the markers.
- Add hover tooltips to show country names.
- Add buttons to switch between map projections (Natural Earth, Mercator, Orthographic).
- Title: "World Map of Life Expectancy (2007)".

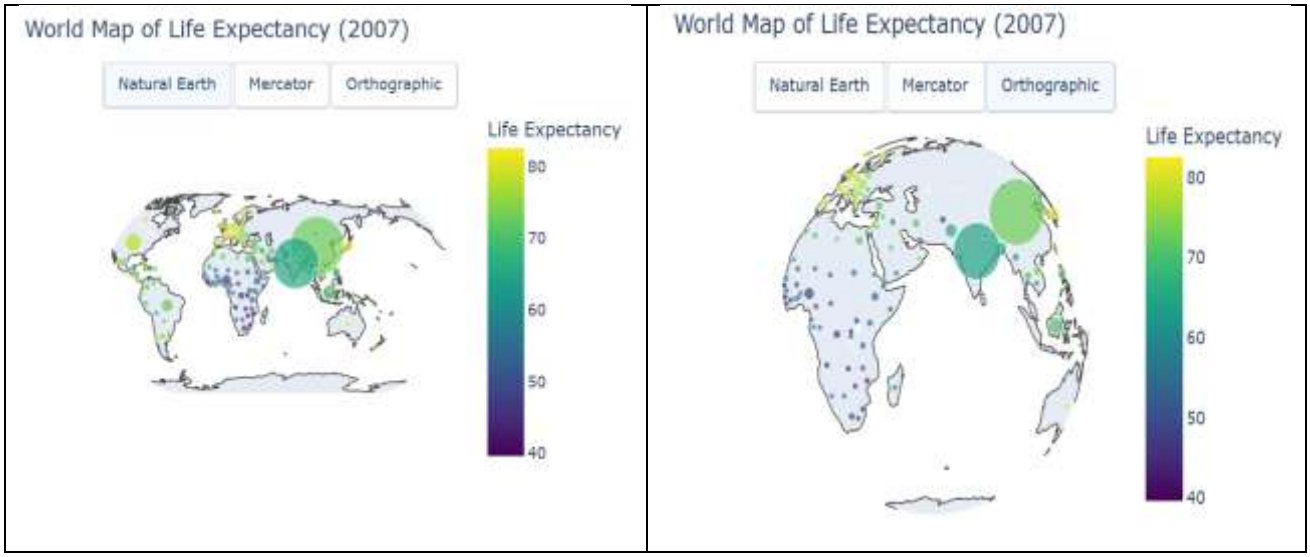
INPUT

```
import plotly.graph_objects as go
import plotly.express as px


data = px.data.gapminder()
```

```
data_2007 = data[data["year"] == 2007]


fig = go.Figure(
    data=go.Scattergeo(
        locations=data_2007["iso_alpha"],
        text=data_2007["country"],
        marker=dict(
            size=(data_2007["pop"] / data_2007["pop"].max()) * 40 + 4,
            color=data_2007["lifeExp"],
            colorscale="Viridis",
            colorbar=dict(title="Life Expectancy"),
            line=dict(width=0.5, color="white")
        ),
        hovertemplate="Country: %{text}<br>Life Exp: %{marker.color:.2f}<br>Population:
%{marker.size:.1f}<extra></extra>"
    )
)


fig.update_layout(
    title="World Map of Life Expectancy (2007)",
    geo=dict(showframe=False, showcoastlines=True, projection_type="natural earth"),
    updatemenus=[
        {
            "type": "buttons",
            "direction": "right",
            "x": 0.5,
            "y": 1.12,
            "xanchor": "center",
            "buttons": [
```

```
            {"label": "Natural Earth", "method": "relayout", "args": [{"geo.projection.type": "natural earth"}]},
            {"label": "Mercator", "method": "relayout", "args": [{"geo.projection.type": "mercator"}]},
            {"label": "Orthographic", "method": "relayout", "args": [{"geo.projection.type": "orthographic"}]}
        ]
    }
  ]
)

fig.show()
```

OUTPUT



TASK 5

Animated Scatter Plot with Play/Pause Controls

- Create an animated scatter plot of GDP per capita vs life expectancy across years.
- Use population (pop) for point size.
- Color points by continent.
- Add play/pause controls to animate automatically.
- Title: "Global Development Trends (1952–2007)".

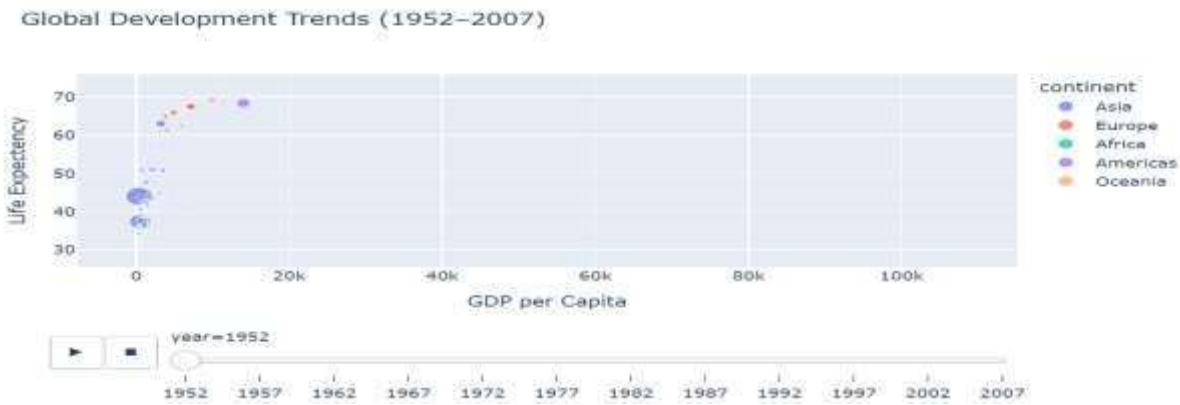
- Label axes appropriately.

INPUT

fig = px.scatter(data, x="gdpPercap", y="lifeExp", title='Global Development Trends (1952–2007)',size="pop", labels={'gdpPercap': "GDP per Capita", 'lifeExp': "Life Expectency"}, animation_frame='year', hover_name='country', color='continent')

fig.show()

OUTPUT



TASK 6

3D Scatter Plot

- Plot GDP per capita (gdpPercap), life expectancy (lifeExp), and population (pop) in 2007.

- Use continent for color coding.

- Size points by population.

- Title: "3D Scatter: GDP, Life Expectancy, Population (2007)".

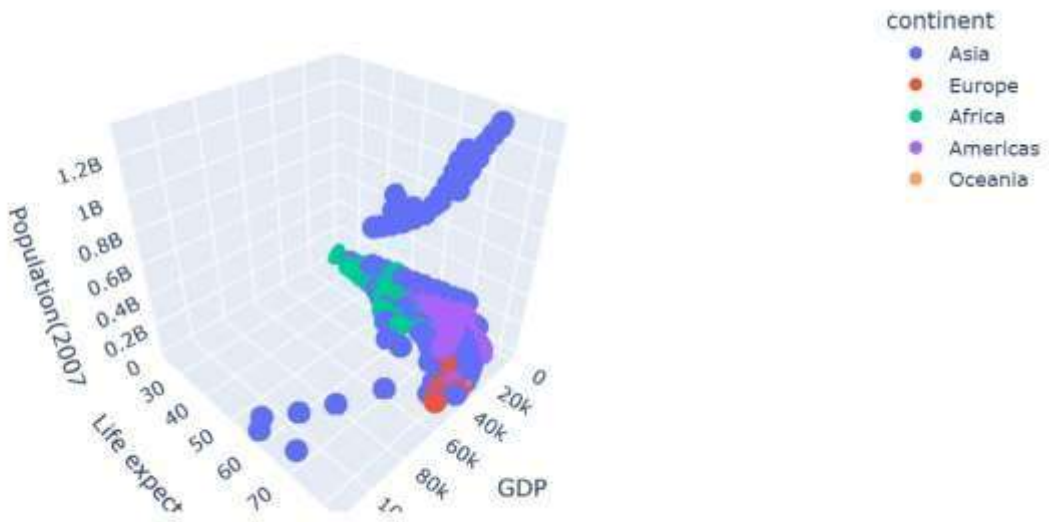
- Label axes appropriately.

INPUT

fig = px.scatter_3d(data, x='gdpPercap', y='lifeExp', z='pop', color='continent', labels={'gdpPercap': 'GDP', 'lifeExp': "Life expectency", 'pop' : "Population(2007"})

fig.show()

OUTPUT



TASK 7

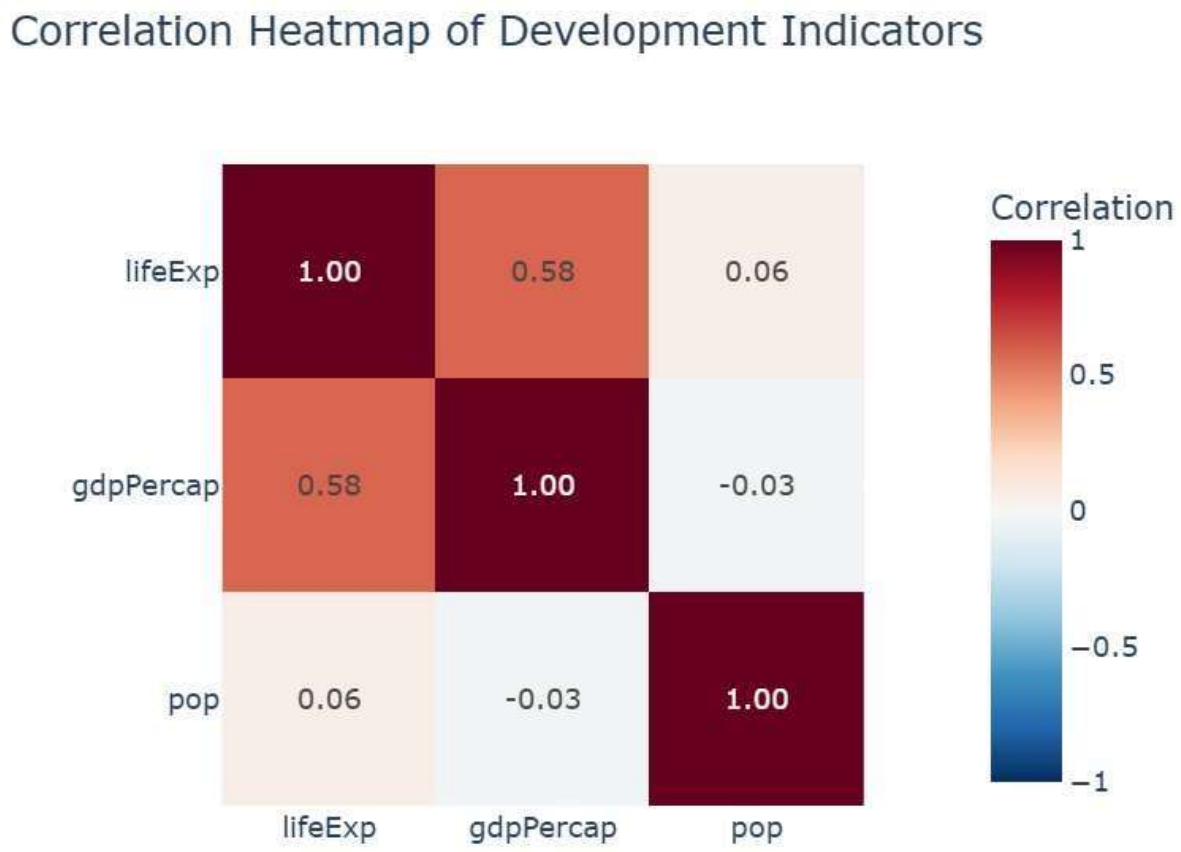
Heatmap

- Compute the correlation matrix for lifeExp, gdpPercap, and pop.
- Plot the matrix as a heatmap.
- Add numeric values inside each cell.
- Title: "Correlation Heatmap of Development Indicators".
- Use a diverging color scale for better contrast.

INPUT

```
corr_matrix = data[["lifeExp", "gdpPercap", "pop"]].corr()


fig = px.imshow( corr_matrix,   text_auto=".2f",
        color_continuous_scale="RdBu_r", zmin=-1, zmax=1,
        title="Correlation Heatmap of Development Indicators",
        labels={"color": "Correlation"})


fig.show()
```

OUTPUT



TASK 8

Pie Chart

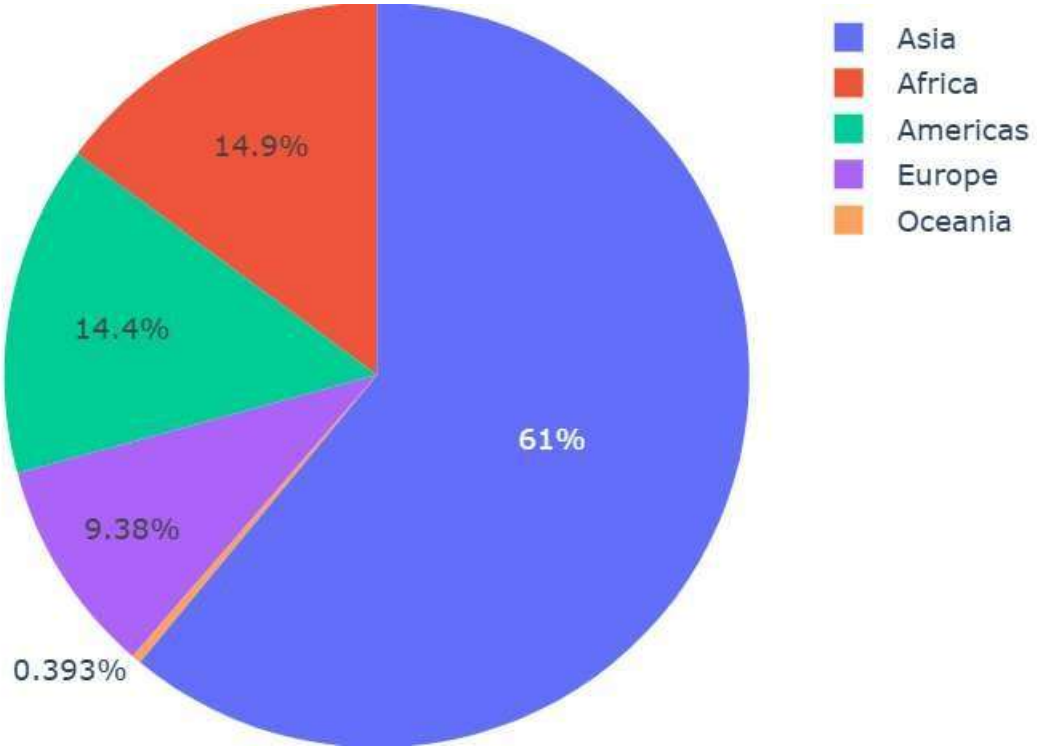
- Create a pie chart showing population share (pop) by continent in 2007.
- Add percentage labels to each slice.
- Title: "Population Share by Continent (2007)".
- Use contrasting colors for slices.

INPUT

```
data_2007 = data[data.year==2007]


pie_plot = px.pie(
data_2007,
names="continent",
values="pop"
)
```

```
pie_plot.show()
```

OUTPUT



TASK 9

Treemap of Population

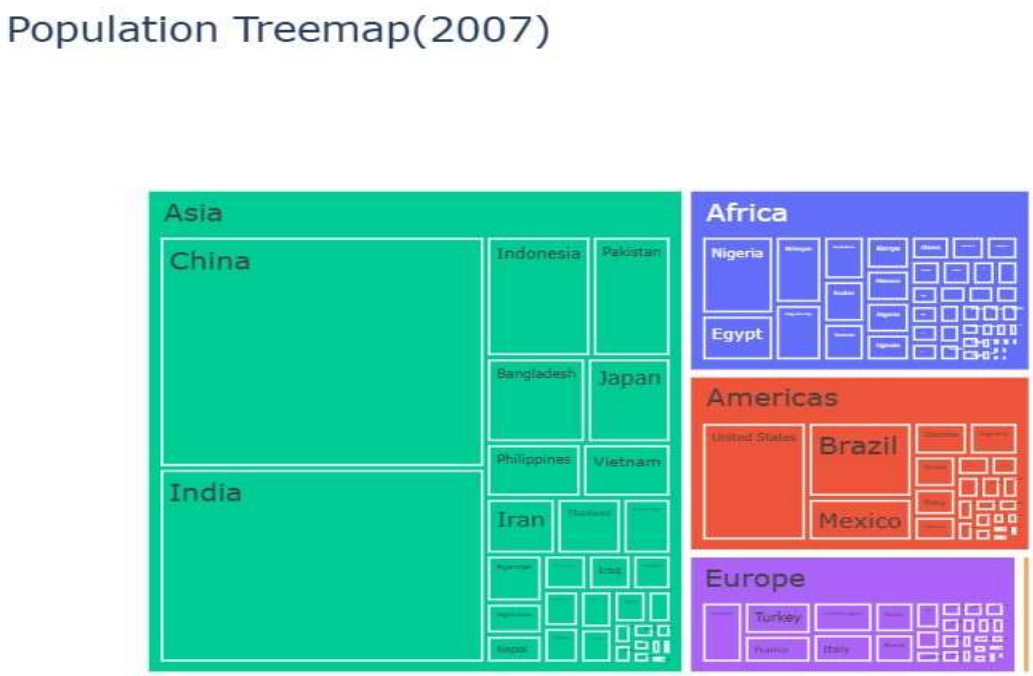
- Title: "Population Treemap (2007)"
- Size blocks by population.
- Color blocks by continent.
- Hover tooltips should show country names.

INPUT

```
treemap_plot = px.treemap(
   data_2007,    path=["continent",
"country"],    values="pop",
color="continent",
hover_name="country",
title="Population Treemap(2007)"
```

```
)
treemap_plot.show()
```

OUTPUT



TASK 10

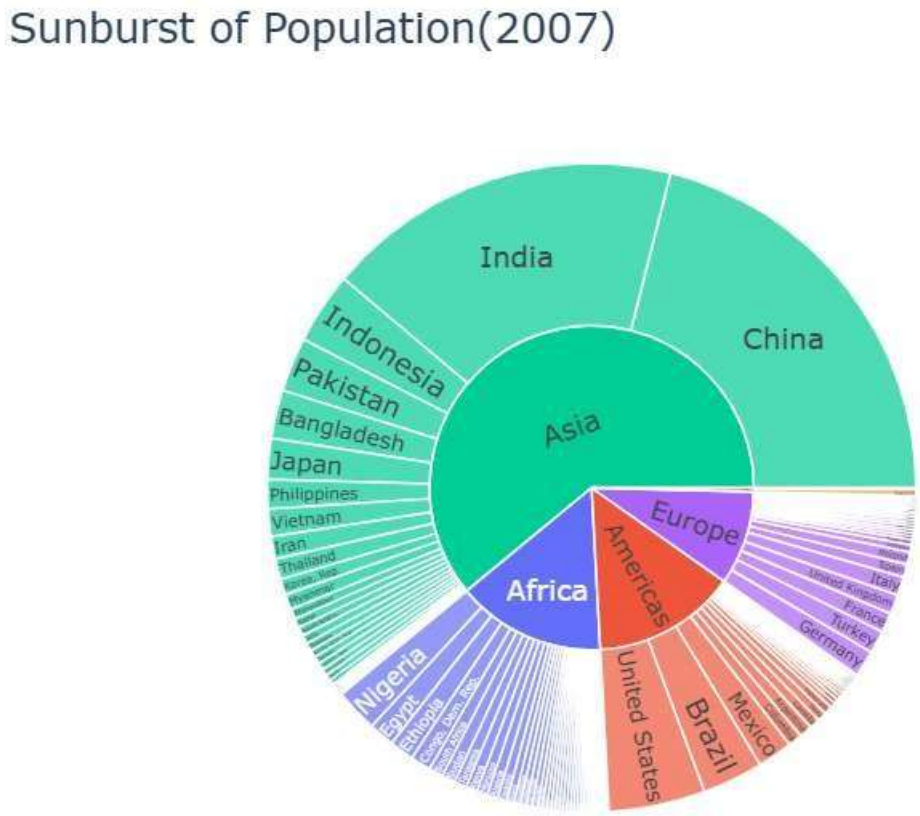
Sunburst Chart

- Title: "Sunburst of Population (2007)".
- Inner ring = continent, outer ring = country.

INPUT

```
sunburst_plot = px.sunburst(
data_2007,    path=["continent",
"country"],    values="pop",
color="continent",
hover_name="country",
title="Subburst of Population(2007)"
)
sunburst_plot.show()
```

OUTPUT


Sunburst of Population(2007)
India
China
Indonesia
Pakistan
Bangladesh
Japan
Philippines
Vietnam
Iran
Thailand
Asia
Europe
Africa
Americas
Italy
France
Turkey
Germany
Nigeria
Egypt
Ethiopia
United States
Brazil
Mexico